Nora Wolcott, PhD

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Education

PhD in Molecular, Cellular, and Developmental Biology
University of California, Santa Barbara *2019 – 2025*
BS in Cell and Molecular Biology; BA in Music Performance
The George Washington University *2015 – 2019*

Professional Experience

Postdoctoral Fellow; Sandeep Datta Lab *07/2025 – present*
Harvard Medical School, Boston, MA

- Worked to implement complex AI-driven neurophysiological and transcriptomic data analysis (scRNA-seq, bulk seq), focusing on steroid-driven modulation in sensory circuits.
- Secured $500k in independent federal and institutional funding.
- Collaborated with external biotech partners to develop and implement multiphoton technologies.

PhD Candidate; Michael Goard Lab *09/2019 – 06/2025*
UC Santa Barbara, Santa Barbara, CA

- Engineered technologies to interrogate learning and memory circuits underlying spatial navigation.
- Used multiphoton microscopy to investigate neural plasticity *in vivo.*
- Investigated the role of steroid hormones in the hippocampus using a custom surgically implanted microprism for longitudinal imaging.
- Built a GUI-based machine learning platform for classification of hormonal state.

Laboratory Assistant; Michael Krashes Lab *06/2018 – 07/2019*
National Institutes of Health NIDDK, Bethesda, MD

- Worked to identify the genetic pathways that govern hunger and satiety.
- Coordinated an optogenetic paradigm for identifying hypothalamic consummatory drivers.

Research Assistant; Arnaud Martin Lab *01/2016 – 06/2019*
The George Washington University, Washington, DC

- Studied how Hox gene patterning influences development in butterflies using CRISPR/Cas9.

Undergraduate Research Assistant; Russell Snell Lab *01/2018 – 06/2018*
University of Auckland, Auckland, NZ

- Used RNA seq to investigate the genetic basis of Huntington's disease.

Honors and Awards

Foundation for Women's Wellness Fellowship Award, FWW	*11/2026*
F99/K00 Advancement in Neuroscience Award, NIH NINDS	*06/2024, 05/2026*
Brooks Fellowship Award, HMS	*04/2026*
Lancaster Dissertation Award, UCSB	*05/2025*
Emerging Scholars in Women's Health Research Award, SWHR	*03/2025*
Elizabeth Young New Investigator Award, OSSD	*02/2023*
Individual Professional Skills Grant awardee, UCSB	*11/2022*
MCDB Research Symposium Best Short Talk Award, UCSB	*10/2022*
Trainee Professional Development Award, Society for Neuroscience	*10/2021*
MCDB Graduate Merit Fellowship, UCSB	*09/2019*
Presidential Academic Scholar, GWU	*08/2015*
Harlan Undergraduate Research Fellow, GWU	*01/2017, 05/2017*
Presidential Scholar in the Arts, GWU	*08/2015*

Outreach and Mentorship

Undergraduate Research Assistant Mentor	*01/2021 – Present*
Teaching Assistant	*01/2019 – 12/2023*
• GWU: Genetic Engineering Laboratory	*01/2019 – 06/2019*
• UCSB: Introduction to Biology Laboratory and Neurobiology	*09/2019 – 12/2023*
Women in Science and Engineering (WiSE) Mentor	*09/2020 – 06/2025*
LGBTQ in STEM Mentor	*01/2021 – 06/2025*
Graduate Representative, Resource Center for Sexual and Gender Diversity	*08/2023 – 06/2025*

Science Communication

Delivered **15+ presentations** to diverse audiences, translating complex data into high-impact narratives.

ThorLabs Multiphoton Imaging Symposium Speaker	*10/2026*
UCSB Donors of Graduate Division Speaker	*10/2025*
Night Lizard Brewing Company Science Pub	*07/2024*
MCDB/BMSE 25th Annual Symposium Invited Speaker	*09/2022*
The Scientist Featured Correspondent	*10/2021*
Lemelson Institute Biogen-MIT Series Speaker	*06/2021*
UCSB Grad Slam Runner Up	*03/2021*

Skills

Medical & Strategic: Scientific Data Translation, Cross-Functional Leadership, KOL/Expert Engagement

Data Science & Informatics: Python, R, MATLAB, Machine Learning, SQL Analytics, Bioinformatic Pipelines, Git Version Control

Laboratory & Translational: Bulk/Single-Cell RNA Sequencing, Next-Generation Sequencing, Immunoassays (ELISA, qPCR), Neurophysiology (electrical, optical), CRISPR-Cas9 Technologies

Publications

Wolcott, N., Redman, W., Karpinska, M., Jacobs, E. & Goard, M. The estrous cycle modulates hippocampal spine dynamics, dendritic processing, and spatial coding. *Neuron* (2025).

Wolcott, N., Sit, K., Raimandi, G., Hodges, T., Shansky, R., Galea, L., Ostroff, L. & Goard, M. Automated classification of estrous stage in rodents using deep learning. *Scientific Reports* (2022).

Redman, W., **Wolcott, N.,** Montelisciani, L., Luna, G., Marks, T., Sit, K., Yu, CH., Smith, S. & Goard, M. Long-term transverse imaging of the hippocampus using implanted microperiscopes. *eLife* (2022).

Wolcott, N. Opinion: Neuroscientists Need to Think About Sex (Bias). *The Scientist* (2021).

Mazzone, C., Lian-Guallpa, J., Li, C., **Wolcott, N.**, et al. High fat food biases hypothalamic and mesolimbic expression of consummatory drives. *Nature Neuroscience* (2020).

Martin, A.. **Wolcott, N.** & O'Connell, L. Bringing immersive science to undergraduate laboratory courses using CRISPR gene knockouts in frogs and butterflies. *Journal of Experimental Biol*ogy (2020).

Oral Presentations

Wolcott, N., Jacobs, E. & Goard, M. (2025, June). *Dynamic endocrine factors modulate hippocampal structure and function.* Society for Women's Health Research Symposium, Albuquerque, NM.

Wolcott, N., Jacobs, E. & Goard, M. (2023, May). *Chronic recording of hippocampus reveals sex differences in spatial navigation circuitry.* Organization for the Study of Sex Differences, Calgary, Alberta, CA.

Wolcott, N., Redman, W., Sit, K., Goard, M. (2022, October). *Investigating the role of steroid hormones in hippocampal plasticity.* MCDB/BMSE 25th Annual Symposium, Santa Barbara, CA.

Wolcott, N. (2021, June). *Ways We Study Alzheimer's.* Lemelson Institute Biogen-MIT Biotech in Action Summer Series, Massachusetts Institute of Technology, Boston, MA.

Poster Presentations

Wolcott, N., Redman, W., Jacobs, E. & Goard, M. (2024, October). *The estrous cycle modulates hippocampal spine dynamics and spatial coding.* Society for Neuroscience annual meeting, Chicago, IL.

Wolcott, N., Redman, W. & Goard, M. (2024, March). *Hippocampal spatial representations are modulated by cyclic endocrine factors.* CoSyNe, Lisbon, Portugal.

Wolcott, N., Jacobs, E. & Goard, M. (2023, May). *Chronic recording of hippocampus reveals sex differences in spatial navigation circuitry.* Organization for the Study of Sex Differences Annual Meeting, Calgary, Alberta, CA.

Wolcott, N., Goard, M. (2023, March). *Dynamic endocrine factors shape hippocampal spatial representations.* CoSyNe, Montreal, Quebec, CA.

Wolcott, N., Redman, W., Sit, K., Goard, M. (2021, November). *Structural plasticity of apical dendritic spines in hippocampal CA1 is modulated by sex-specific steroid hormones.* Society for Neuroscience annual meeting, Chicago, IL.

Wolcott, N., Martin, A., Day, C. (2017, August). *CRISPR-mediated Hox gene knockout in Vanessa cardui: the anteroposterior functions of Ubx and Abd-a*. Wilbur V. Harlan Poster Session, The George Washington University, Washington, DC.